

DIPENDRA SHARMA

Software Development Engineer Intern — Backend & Algorithms

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Summary

Backend-focused developer applying competitive programming instincts to production engineering. Implements DSA concepts directly in shipped code — stack-based state engines, hash-map authorization layers, and complexity-aware query design. Ramps quickly with measurable proof: zero to live deployed systems in under three months, backed by public commit history. Debugs from first principles, tracing root causes through system behavior rather than patching symptoms. Builds effectively under constraints, delivering production-grade software on student resources without hand-holding. Currently expanding into advanced algorithms, system design, and distributed systems.

Problem Solving & Competitive Programming

- **LeetCode:** Peak rating **1570**; 313 problems solved (70.23% acceptance). 365-Day Badge, 100-Day Badge 2026, 361 active days, 130-day max streak, 599+ submissions.
- **Codeforces:** 40+ practice problems solved; actively preparing for first rated contest.
- **Core Topics:** Arrays, Strings, Hash Tables, Two Pointers, Sliding Window, Binary Search, Linked Lists, Stacks, Queues, Trees, Graphs, Heaps, Recursion, Sorting. **Advanced:** DP (30), Divide & Conquer (17), Backtracking (13), Greedy.
- **Production Application:** Designs systems with Big-O awareness — O(1) stack operations for state history, hash-map based RBAC lookups, and index-optimized database queries.

Technical Skills

Languages: C++ (primary for CP), JavaScript, TypeScript, SQL, C

DSA & Problem Solving: Arrays, Trees, Graphs, Heaps, DP, Greedy, Backtracking, Two Pointers, Sliding Window, Binary Search, Stacks, Queues

Backend: Node.js, Express, REST APIs, JWT, RBAC, PostgreSQL, Prisma, MongoDB

System Design: Auto-scaling, scalability, CDN, latency optimization, load balancing (familiar)

Frontend: React, Next.js, TypeScript, Redux, Context API, Tailwind CSS

DevOps & Cloud: Docker, Git, GitHub Actions, AWS (EC2/S3/IAM), Vercel, Render, Linux

Projects

Yeti Jobs | [Github](#) | [Live](#) Jan 2026 – Apr 2026

- Architected **50+ REST APIs** with RBAC — implemented hash-map based role validation for O(1) permission checks across admin, recruiter, and candidate workflows.
- Optimized PostgreSQL via composite indexing leveraging tree-structured search behavior, cutting query time **30%** (7ms → 5ms per EXPLAIN ANALYZE); connection pooling managed **20 concurrent DB connections**.
- Load-tested with Apache Bench: **1,000 requests** at **100 concurrent users** → **67 req/sec**, p95 latency **1,929ms**, **0 failures** on a single free-tier Render instance.
- Multi-stage Docker builds reduced backend image **1.99GB** → **520MB** and frontend **1.03GB** → **220MB** (avg. **76% reduction**); cron job eliminated cold starts from **50s** → **<2s**, sustaining **99.5% uptime** over 3 months.
- 28+ integration tests (Jest + Supertest); deployed via **Vercel** and **Render** with **GitHub Actions** CI/CD and Express rate limiting.

– **Tech:** PostgreSQL, Node.js, Express, React, JavaScript, Docker, JWT, Jest, Supertest, GitHub Actions

State Flow | [Github](#) | [Live](#) Oct 2025 – Jan 2026

- Built Jira-style task manager with custom undo/redo engine using **three-stack state history** (past/present/future) — O(1) push/pop handling **50+ sequential operations** without lag or memory bloat.
- Persisted state via React Context API + localStorage; analytics dashboards (Chart.js) and CSV/JSON export.
- **Tech:** Next.js, TypeScript, React.js, Context API, Tailwind CSS, DnD Kit, Chart.js

Achievements

- LeetCode peak rating **1570** — 313 problems solved, 365-Day Badge, 100-Day Badge 2026.
- Winner — **CodeaThon 7.0** (cash prize, 120+ participants).
- Selected Contributor — **GirlScript Summer of Code 2026**.
- Led multiple hackathon teams of 3-4 members — shipped working prototypes under 24-hour deadlines.
- Active open-source contributor — **20+ GitHub repos**, **400+ commits**.
- **CEED Member** — Entrepreneurship Program, Chitkara University.
- **AWS Educate** — EC2, S3, IAM fundamentals.

Education

Chitkara University

Bachelor of Computer Applications (BCA)

Aug 2024 – Expected 2027

CGPA: 9.44/10

- Coursework: DSA, Computer Networking, Operating Systems, DBMS, Software Engineering, Web Performance